

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026
Homework #5 Solutions

1. (a)

$$\begin{aligned}
 E(X) &= \int_{x=-\infty}^{\infty} x f_X(x) dx \\
 &= \int_{x=-2}^3 \frac{3}{35} x^3 dx \\
 &= \left. \frac{3}{140} x^4 \right|_{x=-2}^{x=3} \\
 &= \frac{243}{140} - \frac{48}{140} \\
 &= \frac{195}{140} \\
 &= \frac{39}{28} \\
 &= 1.393
 \end{aligned}$$

(b) We can compute $\text{Var}(X)$ either as $E\left(X - \frac{39}{28}\right)^2$ or $E(X^2) - \left(\frac{39}{28}\right)^2$. The integration for the latter is a little easier.

$$\begin{aligned}
 E(X^2) &= \int_{x=-\infty}^{\infty} x^2 f_X(x) dx \\
 &= \int_{x=-2}^3 \frac{3}{35} x^4 dx \\
 &= \left. \frac{3}{175} x^5 \right|_{x=-2}^{x=3} \\
 &= \frac{729}{175} + \frac{96}{175} \\
 &= \frac{825}{175} \\
 &= \frac{33}{7} \\
 &= 3.617
 \end{aligned}$$

$$\text{So } \text{Var}(X) = \frac{33}{7} - \left(\frac{39}{28}\right)^2 = 2.774.$$

(c)

$$\begin{aligned}
 E(\cos(X^3)) &= \int_{x=-\infty}^{\infty} (\cos(x^3)) f_X(x) dx \\
 &= \int_{x=-2}^3 \frac{3}{35} x^2 \cos(x^3) dx \\
 &= \left. \frac{1}{35} \sin(x^3) \right|_{x=-2}^{x=3} \\
 &= \frac{1}{35} (\sin(27) - \sin(-8)) \\
 &= 0.0556
 \end{aligned}$$

Note that here 27 and -8 must be interpreted as radians, not degrees- be careful!

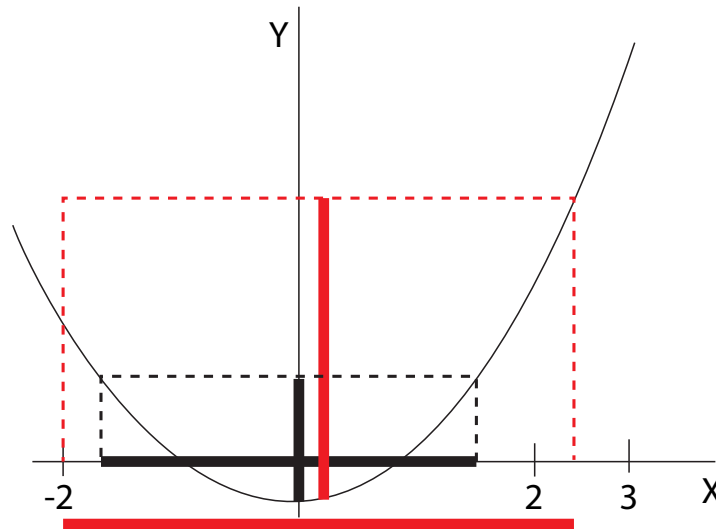
2. (a) This is easy using the properties of expected value:

$$\begin{aligned}
 E(Y) &= E(7X^2 - 1) \\
 &= 7E(X^2) - 1 \\
 &= 33 - 1 \\
 &= 32
 \end{aligned}$$

- (b) First we need the CDF of X , which we can compute as:

$$F_X(x) = \begin{cases} 0 & x < -2 \\ \frac{1}{35}x^3 + \frac{8}{35} & x \in [-2, 3] \\ 1 & x > 3 \end{cases}$$

Now we can compute the CDF of $Y = 7X^2 - 1$. First, since X can take on values in $[-2, 3]$, Y can take on values in $[-1, 62]$. However, one thing to note is that values of Y in the range $[-1, 27]$ can be produced by two values of X (positive or negative) while values of Y in the range $(27, 62]$ can be produced by only one value of X (between 2 and 3). So the CDF will be piecewise. The below sketch shows the issue: the black range of Y maps to a symmetric interval in X , but the red range of Y maps to an asymmetric interval in X (bounded by -2 on the left).



Let's take the range $Y \in [-1, 27]$ first; in this range,

$$\begin{aligned}
 F_Y(y) &= P(Y \leq y) \\
 &= P(7X^2 - 1 \leq y) \\
 &= P\left(X \in \left[-\sqrt{\frac{y+1}{7}}, \sqrt{\frac{y+1}{7}}\right]\right) \\
 &= F_X\left(\sqrt{\frac{y+1}{7}}\right) - F_X\left(-\sqrt{\frac{y+1}{7}}\right) \\
 &= \frac{1}{35}\left(\frac{y+1}{7}\right)^{\frac{3}{2}} + \frac{8}{35} + \frac{1}{35}\left(\frac{y+1}{7}\right)^{\frac{3}{2}} - \frac{8}{35} \\
 &= \frac{2}{35}\left(\frac{y+1}{7}\right)^{\frac{3}{2}}
 \end{aligned}$$

In the range $Y \in (27, 63]$, the CDF will be

$$\begin{aligned}
 F_Y(y) &= P(Y \leq y) \\
 &= P(7X^2 - 1 \leq y) \\
 &= P\left(X \leq \sqrt{\frac{y+1}{7}}\right) \\
 &= F_X\left(\sqrt{\frac{y+1}{7}}\right) \\
 &= \frac{1}{35} \left(\frac{y+1}{7}\right)^{\frac{3}{2}} + \frac{8}{35}
 \end{aligned}$$

Thus the complete CDF of Y is

$$F_Y(y) = \begin{cases} 0 & y < -1 \\ \frac{2}{35} \left(\frac{y+1}{7}\right)^{\frac{3}{2}} & y \in [-1, 27] \\ \frac{1}{35} \left(\frac{y+1}{7}\right)^{\frac{3}{2}} + \frac{8}{35} & y \in [27, 62] \\ 1 & y > 62 \end{cases}$$

3. (a) The Markov bound says that $P(X \geq 25) \leq \frac{22}{25} = 0.88$.
 - (b) The Chebyshev bound says that $P(|X - 22| \geq 3) \leq \left(\frac{\sqrt{2}}{3}\right)^2 = 0.22$, which, assuming the distribution is symmetric, would bound $P(X \geq 25) \leq 0.11$ (a lot less than the previous estimate).
 - (c) If the random variable is in fact Gaussian, we can compute from the Q table that $P(X \geq 25) = Q\left(\frac{3}{\sqrt{2}}\right) = Q(2.12) = 0.017$. So while both bounds are satisfied, neither is very tight. (The bounds work better when we ask for values really far from the mean.)
4. (a) First we note that the PDF of R is given by

$$f_R(r) = \begin{cases} 2 & r \in [1.5, 2] \\ 0 & \text{otherwise} \end{cases}$$

If the waterslide is 64m long, the volume V and radius R are related by $V = 32\pi R^2$. Let's call this function $g(R)$. Since R is positive, this is a one-to-one function and we can also write $g^{-1}(V) = \left(\frac{V}{32\pi}\right)^{\frac{1}{2}}$. We can compute that the minimum and maximum values of the waterslide volume are 72π and 128π respectively. While we could go through the procedure similar to the above by determining the CDF of V and then taking the derivative, we can also apply the direct formula to get the PDF:

$$\begin{aligned}
f_V(v) &= f_R(g^{-1}(v)) \cdot \frac{d}{dv} g^{-1}(v) \\
&= f_R\left(\left(\frac{v}{32\pi}\right)^{\frac{1}{2}}\right) \cdot \frac{1}{64\pi} \left(\frac{v}{32\pi}\right)^{-\frac{1}{2}} \\
&= \begin{cases} \frac{1}{32\pi} \left(\frac{v}{32\pi}\right)^{-\frac{1}{2}} & \left(\frac{v}{32\pi}\right)^{\frac{1}{2}} \in [1.5, 20] \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} \frac{1}{\sqrt{32\pi}\sqrt{v}} & v \in [72\pi, 128\pi] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

- (b) Note that $W = \left(\frac{V}{\pi}\right)^{\left(\frac{1}{2}\right)}$, but plugging in V in terms of R , we get the easier expression $W = 4\sqrt{2}R$. We can go through the same type of one-to-one process above, or we can immediately see that W will also be a uniform random variable, just over a wider range:

$$f_W(w) = \begin{cases} \frac{1}{2\sqrt{2}} & w \in [6\sqrt{2}, 8\sqrt{2}] \\ 0 & \text{otherwise} \end{cases}$$